
EE 513: Special Electrical Machines

Course Name: EE 513

Course Code: Special Electrical Machines

L-T-P-C: 3-0-0-3

Prerequisites: EE 201 Electromechanics

Students intended for: UG/PG

Elective or Compulsory: Elective

Approval: 8th Senate

Course Contents

Permanent Magnet Brushless D.C. Motors: Fundamental equations EMF and Torque equations Torque speed characteristics Rotor position sensing Sensorless motors Motion control. [9 Lectures]

Permanent Magnet Synchronous Motors: Construction - Principle of operation EMF and torque equations Starting Rotor configurations Dynamic model. [9 Lectures]

Synchronous Reluctance Motors: Constructional features axial and radial flux motors operating principle characteristics. [8 Lectures]

Switched Reluctance Motors: Constructional features principle of operation torque production characteristics power controllers. [8 Lectures]

Stepping Motors: Features fundamental equations PM stepping motors Reluctance stepping motors Hybrid stepping motors Torque and voltage equations characteristics. [8 Lectures]

Text Books:

1. Miller, T. J. E., **Brushless Permanent Magnet and Reluctance Motor Drives**, Oxford Science Publications, 1989.
2. Kenjo, T., and Sugawara, A., **Stepping Motors and their Microprocessor Controls**, Oxford Science Publications, 1984.
3. Venkataratnam K., **Special Electrical Machines**, CRC Press, 2009.

Reference Books:

1. Krishnan, R., **Permanent Magnet and BLDC Motor Drives**, CRC Press, 2009.
2. Chang-liang, X., **Permanent Magnet Brushless DC Motor Drives and Controls**, 2012.